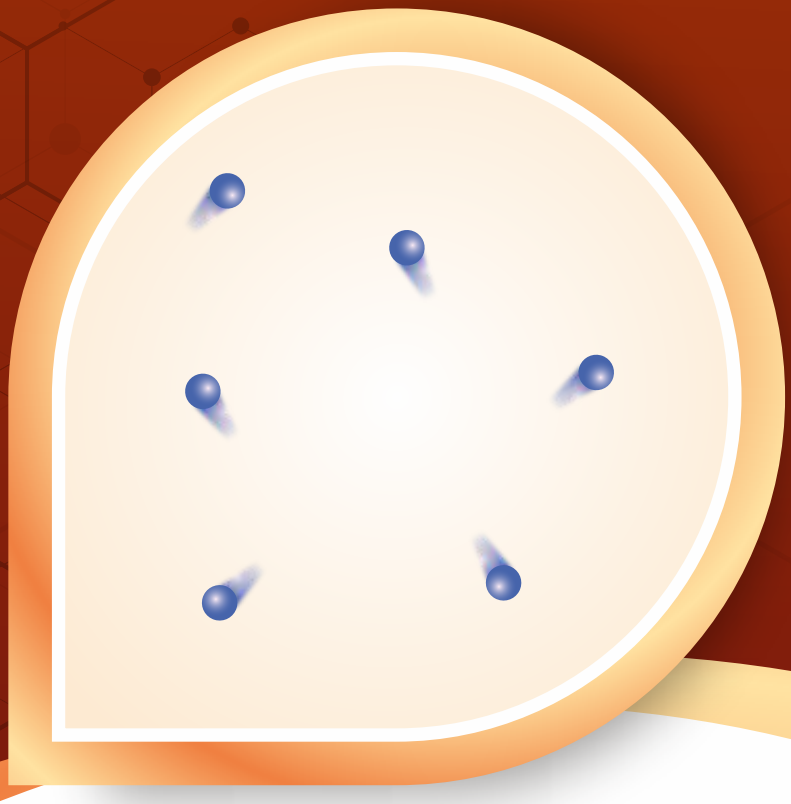
ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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# PHYSICAL CHEMISTRY

## JEE (MAIN+ADVANCED)

NURTURE COURSE



## Exercise

# Ideal Gas

(English Medium)

## EXERCISE # S-I

### BOYLE'S LAW

**1.** Certain mass of a gas occupy 400 ml at 1.5 atm and 27°C. Calculate the volume (in ml) occupied by same mass of the gas at 1.2 atm and 27°C.

**IG0170**

**2.** 1.80 $dm^3$ of a gas is expanded upto 9.0 $dm^3$ at constant temperature. If the final pressure of gas becomes 20 kPa, the initial pressure of gas (in bar) was

**IG0171**

**3.** The pressure of certain mass of a gas is increased by 60% at constant temperature. The percentage decrease in the volume of gas is

**IG0172**

**4.** Two glass bulbs A and B are connected by a very small tube having a stop cock. Bulb A has a volume of 100 ml and contained some ideal gas, while bulb B was empty. On opening the stop cock, the pressure fell down to 40% at constant temperature. Find out the volume of bulb B (in mL).

**IG0004**

### CHARLES LAW

**5.** Calculate the volume (in L) which 4 litre of gas at 0ºC will occupied at 100ºC and the same pressure.

**IG0006**

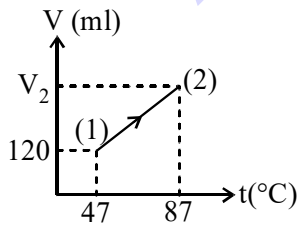
**6.** A certain amount of a gas at 27°C and 1 bar pressure occupies a volume of 25 $m^3$. If the pressure is kept constant and the temperature is raised to 87°C, what will be the volume (in $m^3$) of the gas ?

**IG0008**

**7.** A balloon is filled upto $\frac{3}{4}$ th of its maximum streching capacity at 27°C. The maximum temperature (in °C) upto which the balloon may be heated safely at constant pressure, is.

**IG0173**

**8.**




The value of $V_2$ is.

**IG0174**

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## PRESSURE LAW

**9.** In a rigid closed vessel, the pressure exerted by gas is 57 cm Hg at 40°C. If it is heated to 80°C, the final pressure of gas (in cm Hg) becomes.

**IG0175**

**10.** The temperature (in °C) of certain mass of a gas is increased by 10% at constant volume. If the initial temperature was 100°C, the percentage increase in the pressure of gas is ..... %.

**IG0176**

## AVOGADRO'S LAW

**11.** 10 gm of a gas occupy 123 $cm^3$. Calculate the volume (in $cm^3$) occupied by 2 gm of the same gas at the same temperature and pressure.

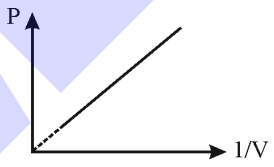
**IG0177**

**12.** 1.20 moles of $CH_4$ gas occupy 38.94 $dm^3$ at certain pressure and temperature. How many moles of $SO_3$ gas will occupy 19.47 $dm^3$ at the same pressure and temperature.

**IG0178**

## IDEAL GAS EQUATION

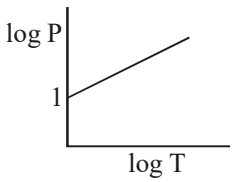
**13.** Consider the following graph




Graph is plotted for 1 mol of gas at 400K, find slope of curve (in L-atm). [Take : $R = 0.08 \frac{L-atm}{mol-K}$ ]

**IG0003**

**14.** In constant volume, container of 0.82 litre, log P vs log T is plotted as shown in graph. Calculate number of moles of ideal gas present in container :



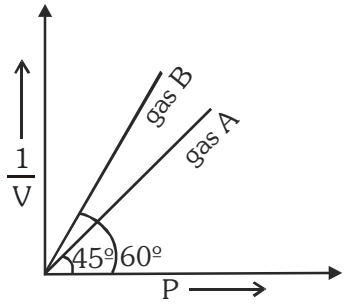

**IG0007**

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**15.** At constant temperature of 273 K, $\left(\frac{1}{V}\right)$ v/s P are plotted for two ideal gases A and B as shown.

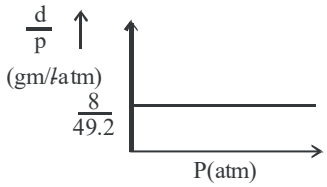
(V in litre & P in atm)




Find the ratio of number of moles of A and B gases. **IG0011**

**16.** For an ideal gas, the following graph is obtained at constant temperature of 300 K.

The molar mass of gas (in gm/mol) is -




**IG0012**

**17.** The density of phosphorus vapours at 327ºC and 1 atm is 2.52 gm/lit. If molecular formula of phosphorus is $P_X$, then calculate 'x'. **(Atomic weight of : P = 31)**

**IG0013**

**18.** Density of ideal gas at 2.46 atm and 300 K is 0.8 gm/L. Hence molar mass of gas (in gm $mol^{-1}$) is.

**IG0014**

**19.** In a hospital respiratory unit, the pressure gauge reads 4500 mm Hg for 10 L tank containing compressed $O_2$. How many litres of $O_2$ can be delivered from this tank at atmospheric pressure of 750 mm Hg.

(Take temperature to be constant at TK) **IG0015**

**20.** A human adult breathes in approximately 0.50 $dm^3$ of air at 1.00 bar with each breath. If an air tank holds 100 $dm^3$ of air at 200 bar, how many breathes (in order of $10^4$) the tank will supply ?

**IG0017**

**21.** A gas at a pressure of 5.0 bar is heated from 0°C to 546°C and simultaneously compressed to one third of its original volume. What will be the final pressure (in bar) ?

**IG0018**

**22.** 3.6 g of an ideal gas was injected into a bulb of internal volume of 8 L at pressure P atm and temp T K. The bulb was then placed in a thermostat maintained at (T + 15)K, 0.6 g of the gas was let off to keep the original pressure. The value of 'P' is **(Molar mass of gas = 41 gm $mol^{-1}$)**

**G0019**

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**23.** While resting, the average human male use 0.2 $dm^3$ of $O_2$ per hour at 1 atm & 300 K for each kg of body mass. Assume that all this $O_2$ is used to produce energy by oxidising glucose in the body . (i) What is the mass (in gm) of glucose required per hour by a resting male having mass 60 kg . (ii) What volume (in $dm^3$), at 1 atm & 300 K of $CO_2$ would be produced ($R = 0.08 \frac{L-atm}{mol-K}$)

**IG0016**

## DALTON'S LAW OF PARTIAL PRESSURE

**24.** The partial pressures of $N_2$, $O_2$ and $CO_2$ in a vessel are 38 cm of Hg, 190 torr and 0.5 atm, respectively . The total pressure of the mixture (in atm) at the same temperature is.

**IG0020**

**25.** Equal masses of ethane and hydrogen are mixed in an empty container at 25ºC. The percentage of the total pressure exerted by hydrogen is

**IG0021**

**26.** A mixture of hydrogen and oxygen at one bar pressure contains 20% , by weight of hydrogen. The partial pressure (in bar) of hydrogen will be

**IG0022**

**27.** A closed vessel contains equal number of nitrogen and oxygen molecules at pressure of 38.96 mm. If nitrogen is removed from the system, then the pressure (in mm) will be

**IG0023**

**28.** If 2 lit. of gas A at 1.5 atm and 3 lit. of gas B at 2 atm are mixed in a 5 lit. container then find the final pressure (in atm), considering all are at same temperature.

**IG0024**

**29.** One mole of $N_2$ and 3 moles of $H_2$ are taken in a container of capacity 8.2 lit. at 300 K to produce $NH_3$. Find the partial pressures (in atm) of $N_2$, if partial pressure of $NH_3$ after sufficient time was found to be 3 atm.

**IG0025**

**30.** A gaseous mixture at 760 mm in a cylinder has 65% $N_2$, 15% $O_2$ and 20 % $CO_2$ by volume. Calculate the partial pressure (in atm) of $O_2$ gas.

**IG0027**

**31.** The density of a mixture of $O_2$ and $N_2$ gases at 1 atm and 273K is 0.0013 gm/ml. the partial pressure (in atm) of $O_2$ in the mixture is

**IG0029**

**32.** A containers contains air above liquid water. Total pressure was 800 torr. What will be the final pressure (in torr) if volume is doubled. (Aqueous tension = 40 torr)

**IG0030**

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**33.** One litre flask contains air, water vapour and a small amount of liquid water at a pressure of 200 mmHg. If this is connected to another one litre evacuated flask, what will be the final pressure (in mm Hg) of the gas mixture at equilibrium? Assume that temperature to be 50ºC. Aqueous tension at 50°C = 93 mmHg.

**IG0031**

## GRAHAM'S LAW OF DIFFUSION OR EFFUSION

**34.** If helium and methane are allowed to diffuse out of the container under the similar conditions of temperature and pressure, then the ratio of rate of diffusion of helium to methane is x : 1.00. The value of 'x' is

**IG0032**

**35.** 100 ml $SO_2$ gas diffuses in 5 minutes. Under identical conditions, 200 ml of an unknown gas takes 7.5 minutes in diffusion. The molar mass of unknown gas (in gm $mol^{-1}$) is.

**IG0179**

**36.** 4 gm HF gas effuses through a pin hole in a certain time. What mass (in gm) $SO_3$ gas will effuse through the same pin hole, under identical conditions ? (F = 19)

**IG0180**

**37.** A sample of pure Ne gas (M = 20) diffuses $\sqrt{2}$ times faster than a mixture of $CH_4$ and $SO_2$ gases, under identical conditions. The mass percent of $CH_4$ in the mixture of $CH_4$ and $SO_2$ gases is.

**IG0181**

**38.** Under identical conditions, gas 'X' diffuses 1.2 times faster than gas 'Y'. If the density of gas 'Y' is 7.2 gm/litre, the density of gas 'X' in the same conditions (in gm/litre) is.

**IG0182**

**39.** At a particular fixed temperature the gas $A_n$ is 52% dissociated according to the following reaction $A_n(g) \rightleftharpoons nA(g)$

The equilibrium mixture effuses 1.25 times slower than pure oxygen gas under identical condition. If atomic wt. of A is 32, then find **"n"** ?

**IG0033**

**40.** A mixture of $H_2$ and $O_2$ in 2 : 1 mole ratio is allowed to diffuse through a orifice. If the mole ratio of gases coming out initially is $r_{H_2} : r_{O_2} = x : 1$, the value of 'x' is **IG0035**

**41.** $H_2$ and $O_2$ are kept in mass ratio 1 : 8 respectively at 6 atm. If small orifice is made, then moles of $H_2$ with respect to $O_2$ coming out initially, is

**IG0034**

**42.** A mixture of $N_2$ and $H_2$ has initially mass ratio of 196 : 1 then find after how many steps we can obtain a mixture containing 1 : 14 mole ratio of $N_2$ and $H_2$.

**IG0037**

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**43.** The ratio of moles of $SO_2$ to $CH_4$ after fifth diffusion steps is x : 1.00, if their initially mole ratio is 8 : 1. The value of 'x' is. **IG0038**

## KINETIC THEORY OF GASES

**44.** Calculate the average speed (in m/s) of molecules of an ideal gas -

(Molar mass = $\frac{166.28}{\pi}$ gm $mol^{-1}$) at 127°C.

**IG0183**

**45.** Root mean square speed of an unknown gas at 727ºC is $10^5$ cm/second. Calculate molar mass of unknown gas (in gram/mole) [Take R = $\frac{25}{3}$ J/mole-K]. **IG0040**

**46.** How many times is the rms speed of molecules in 8 gm $O_2$ gas at 1200 K and 10 bar, relative to rms speed of molecules in 16 gm $O_2$ gas at 300K and 20 bar ?

**IG0041**

**47.** The most probable speed of molecules of an ideal gas is 300 m/s at 27°C. The molar mass of gas (in gm $mol^{-1}$) is (R = $\frac{25}{3}$ J/K.mol)

**IG0184**

**48.** Calculate the most probable speed (in $10^5$ m/s) of electrons at 127°C, assuming them as isolated gaseous particles. **(Mass of each electron = $9.1 \times 10^{-31}$ kg, $N_A = 6 \times 10^{23}$)**

**IG0185**

**49.** Calculate the total translational kinetic energy (in J) of $CH_4$ gas molecules, present in its 0.8 gm sample, at 27°C)

**IG0186**

**50.** Average translational kinetic energy of an ideal gas molecule at 27ºC is $3.88 \times 10^{-x}$ eV. Hence x is **(1 eV = $1.602 \times 10^{-19}$ J)** **IG0042**

**51.** When the temperature of an ideal gas is increased from 27°C to 927°C, the kinetic energy will be increased by x times. Find the value of x ? **IG0043**

**52.** At what temperature (in °C), the graph between $\left(\frac{1}{du}\cdot\frac{dN}{N}\right)$ (Y-axis) and u (X-axis) will be maximum at 200 $ms^{-1}$ for a sample of ideal gas (molar mass = 41.57 gm $mol^{-1}$) ?

**IG0187**

**53.** A sample of gas contains only 100 molecules in which 40 molecules are travelling at 10 m/s, 35 molecules at 20 m/s and rest, at 30 m/s. The ratio of $u_{rms}$ to $u_{av}$ of molecules is x : 1. The value of 'x' is

[$\sqrt{405}$ = 20.12, $\sqrt{410}$ = 20.25, $\sqrt{415}$ = 20.37, $\sqrt{425}$ = 20.62].

**IG0188**

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## COLLISION PARAMETERS

**54.** If the mean free path is 10 cm at one bar pressure, then its value (in cm) at 5 bar pressure, if temperature is kept constant is

**IG0045**

**55.** Two flask A and B have equal volume at 100 K and 200 K and have pressure 4 atm and 1 atm respectively. The flask A contains $H_2$ gas and B contains $CH_4$ gas. The collision diameter of $CH_4$ is twice that of $H_2$. If ratio of mean free path of $CH_4$ to $H_2$ is 1 : x, the value of 'x' is

**IG0044**

**56.** The temperature of a sample of ideal gas is doubled at constant volume. The collision frequency ($Z_{11}$) of gas molecules becomes ..... times.

**IG0189**

**57.** The temperature of a sample of ideal gas is doubled at constant pressure. The collision frequency ($Z_{11}$) of gas molecules becomes ..... times.

**IG0190**

**58.** The temperature of a sample of ideal gas is increased at constant volume such that the most probable speed of molecules becomes four times. The collision number ($Z_1$) of gas molecules becomes ..... times.

**IG0191**

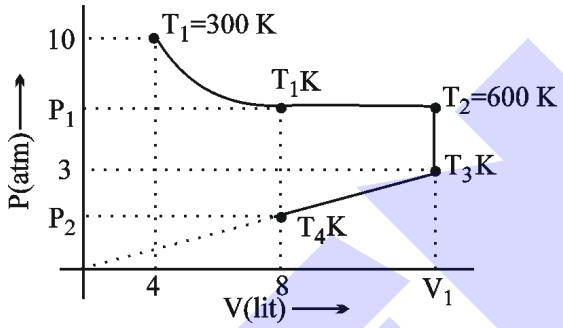
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## EXERCISE # S-II

**1.** The degree of dissociation is 0.4 at 400K and 1 atm for the gaseous reaction :
$2HI(g) \rightarrow H_2(g) + I_2(g)$.
Assume ideal behaviour of all the gases, calcualte the density (in gm/litre) of mixture at 400K and 1 atm **[At. wt. of Iodine = 127 ; Use R = 0.08 atm-litre/K-mole]**

**IG0047**

**2.** Fixed mass of a gas is subjected to the changes as shown is diagram, calculate $T_3$, $T_4$, $P_1$, $P_2$ and $V_1$ as shown is diagram. Considering gas obeys PV = nRT equation.




**IG0048**

**3.** A balloon containing 1 mole air at 1 atm initially is filled further with air till pressure increases to 3 atm. The initial diameter of the balloon is 1 m and the pressure at each state is proportion to diameter of the balloon. Calculate :

**(a)** No. of moles of air added to change the pressure from 1 atm to 3 atm.

**(b)** balloon will burst if either pressure increases to 7 atm or volume increases to $36\pi$ $m^3$. Calculate the number of moles of air that must be added after initial condition to burst the balloon.

**IG0049**

**4.** One mole of $NH_4Cl$ (s) is kept in an open container & then covered with a lid. The container is now heated to 600 K where all $NH_4Cl$ (s) dissociates into $NH_3$ & HCl (g). The Volume of the container is 24.6 litres Assume that outside air is at 300 K and 1 atm pressure.
(i) What will be the final pressure (in atm) of gases inside the container.
(ii) Find whether the lid would stay or bounce off if it can with stand a pressure difference of 5.5 atm.

**IG0050**

**5.** One mole of an ideal gas is subjected to a process in which $P = \frac{1}{8.2} V$ where P is in atm & V in litre. If the process is operating from 1 atm to finally 3 atm (no higher pressure achieved during the process) then what would be the maximum temperature (in °C) obtained.

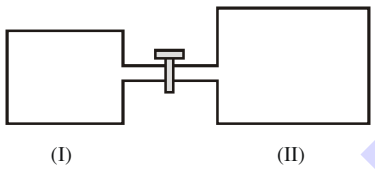
**IG0051**

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**6.** A compound exists in the gaseous state both as a monomer (A) and dimer ($A_2$). The molecular weight of the monomer is 48. In an experiment, 96 g of the compound was confined in a vessel of volume 33.6 litres and heated to 273° C. Calculate the pressure (in atm) developed, if the compound exists as a dimer to the extent of 50 per cent by weight, under these conditions.

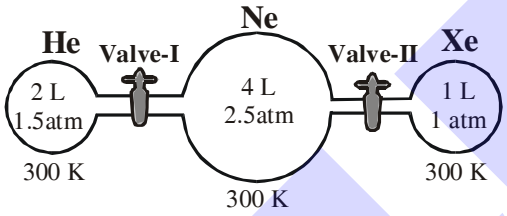
**IG0052**

**7.** Two container (I) & (II) with volume 2 litre & 3 litre are separated through a stopcock as shown in the diagram. Container (I) contains three gases A, B, & C with mole ratio 1 : 2 : 3 and total pressure 30 atm. If stopcock is opened then, calculate the pressure (in atm) of gas C in container (II). (Assume temperature remains constant)




**IG0053**

**8.** For the following diagram, calculate final pressure (in atm) when both valves are opened. Given that temperature (T) is same for each container.




**IG0054**

**9.** At 30ºC dry air [75% $N_2$ + 25% $O_2$] is placed over $H_2O$ (l) at 800 torr combined pressure of all 3 gases. If pressure is gradually increased isothermally to 1560 torr, then calculate partial pressure of $O_2$ at high pressure, in torr. $[V.P_{H_2O} = 40 \text{ torr at } 30^{\circ}C]$.

**IG0055**

**10.** 16 gm of $O_2$ was filled in a container of capacity 8.2 lit. at 300 K. Calculate

**(i)** Pressure exerted by $O_2$ gas, in atm

**(ii)** Total pressure (in atm) of mixture of $O_2$ and $O_3$, if 50 % of oxygen is converted into ozone at same temperature.

**(iii)** Total pressure (in atm) exerted by gases, if 50% of oxygen is converted into ozone ($O_3$) at temperature 50 K.

**IG0056**

**11.** To an evacuated vessel with movable piston under external pressure of 1 atm., 0.1 mol of He and 1.0 mol of an unknown compound (vapour pressure 0.68 atm. at 0°C) are introduced. Considering the ideal gas behaviour, the total volume (in litre) of the gases at 0°C is close to **[JEE 2011]**

**IG0161**

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**12.** Find the number of diffusion steps required to separate the isotopic mixture initially containing some amount of $H_2$ gas and 1 mol of $D_2$ gas in a container of 3 litre capacity maintained at 24.6 atm & 27 °C to the final mass ratio $\left(\frac{w_{D_2}}{w_{H_2}}\right)$ equal to $\frac{1}{4}$. **IG0058**

**13.** The average speed of molecules in a sample of ozone gas is $1.84 \times 10^4$ cm $s^{-1}$. The average translational kinetic energy per molecules is xJ. The value of $\frac{x}{2} \times 10^{22}$ is

(Given : $N_A = 6 \times 10^{23}$ ; $\sqrt{\frac{8}{3\pi}} = 0.92$)

**IG0059**

**14.** $H_2$ gas is kept inside a container A and container B each having volume 2 litre under different conditions which are described below. Determining the missing values with proper unit.

**[R = 8 J $mol^{-1}$ $K^{-1}$ and $N_A = 6 \times 10^{23}$, N = No. of molecules]**

| **Parameter** | **Container A** | **Container B** |
|---|---|---|
| P | (i)– – – – – | 1atm |
| T | 300K | 600 K |
| N | $6\times10^{20}$ | (ii)– – – – – |
| Total Average KE | (iii)– – – – – | (iv)– – – – – |
| Ratio $U_{mps}$ | (v)– – – – – – – – – – – – – | |
| Ratio $Z_{11}$ | (vi)– – – – – – – – – – – – – | |

**IG0060**

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## EXERCISE # O-I

**Single Correct :**

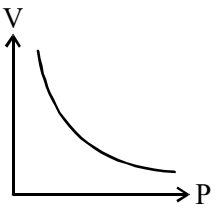
### BOYLE'S LAW

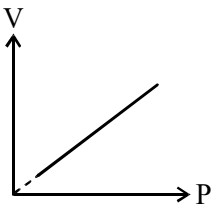
**1.** Certain mass of a gas occupy 222 $cm^3$ at 950 torr. The volume occupied by same mass of gas at same temperature is 333 $cm^3$. The final pressure of gas is :

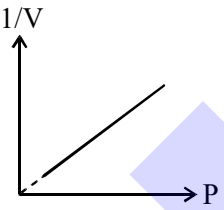
(A) 0.833 atm (B) 1425 torr (C) 1.875 atm (D) 633.33 cm Hg

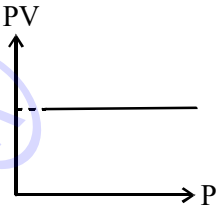
**IG0192**

**2.** Which of the following graph is incorrect according to Boyle's law ?

(A) 

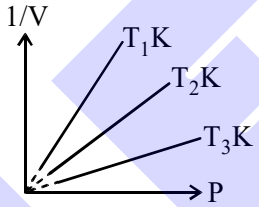
(B) 

(C) 

(D) 


**IG0193**

**3.** The following graph is obtained for a fixed mass of a gas, at different temperatures :




The correct relation between $T_1$, $T_2$ and $T_3$ is :

(A) $T_1 = T_2 = T_3$ (B) $T_1 < T_2 < T_3$ (C) $T_1 > T_2 > T_3$ (D) $T_1 < T_2 > T_3$

**IG0194**

**4.** Which of the following relation is correct for fixed moles of a gas at constant temperatures ?

(A) $\frac{d(\log P)}{d(\log V)} = +ve$ (B) $\frac{dP}{dV} = zero$ (C) $\frac{d\left(\frac{1}{V}\right)}{dP} = -ve$ (D) $P_1.d_2 = P_2.d_1$

(d = density of gas)

**IG0195**

**5.** The volume of fixed mass of a gas is increased by 25%, at constant temperature. For it, the pressure of gas is :

(A) Increased by 25% (B) Decreased by 25%

(C) Increased by 20% (D) Decreased by 20%

**IG0196**

### CHARLES LAW

**6.** Certain mass of a gas occupy 0.4 $m^3$ at 27°C. At what temperature and same pressure, the gas will occupy 0.8 $m^3$ ?

(A) 54°C (B) 600°C (C) 327°C (D) –123°C

**IG0197**

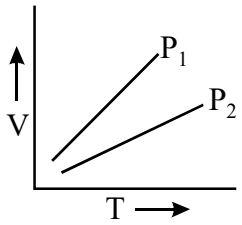
node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\V4-Ideal-Reas-Atomic-Chemical Eq\Eng\1. Ideal\02. Ex.p65

**7.** 10 g of a gas at 1 atm, 273 K occupies 5 litres. The temp. at which the volume becomes double for the same mass of gas at the same pressure is ?

(A) 273K (B) $-273^0$C (C) $273^0$C (D) $546^0$C

**IG0065**

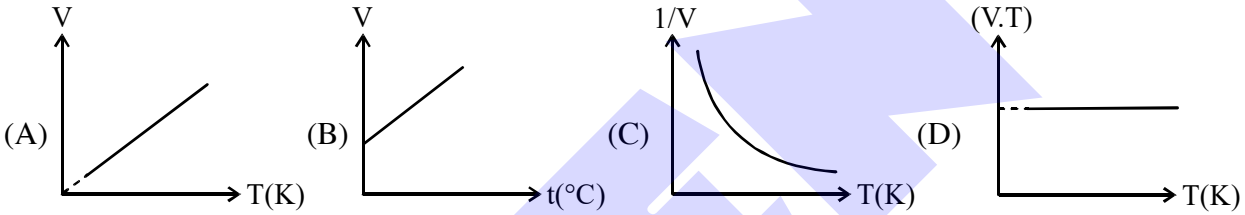
**8.** V versus T curves at constant pressure $P_1$ and $P_2$ for an ideal gas are shown in Fig. Which is correct




(A) $P_1 > P_2$ (B) $P_1 < P_2$ (C) $P_1 = P_2$ (D) All

**IG0067**

**9.** Which of the following graph is incorrect for Charle's law ?




**IG0198**

**10.** The absolute temperature of a sample of gas is increased by 20% at constant pressure. The volume of gas will :

(A) Increase by 20% (B) Decreased by 20%

(C) Increase by 25% (D) Increase by 16.67%

**IG0199**

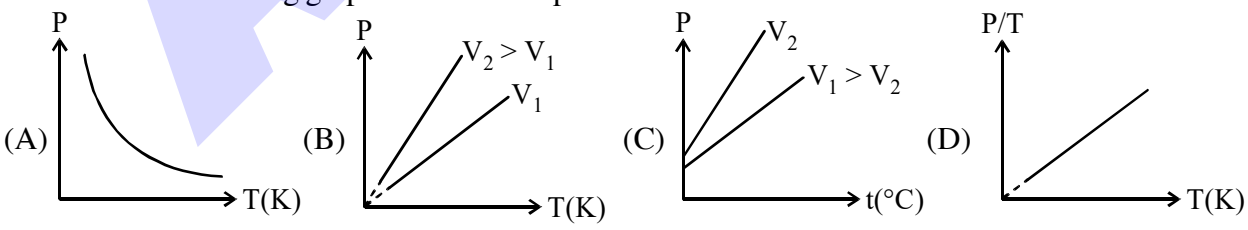
## PRESSURE LAW

**11.** The pressure of a gas in a cylinder is 3.0 atm at 87°C. At what temperature, the pressure of gas will drop to 912 mm Hg ?

(A) 144°C (B) –129°C (C) 417°C (D) 1167°C

**IG0200**

**12.** Which of the following graph is correct for pressure law ?




**IG0201**

**13.** On increasing the temperature (in °C) of certain mass of a gas by 20%, at constant volume, the pressure of gas increased by 5%. The initial temperature of gas is :

(A) 91°C (B) 91 K (C) 364°C (D) –182°C

**IG0202**

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## AVOGADRO'S LAW

**14.** 5.0 gm of a gas occupy 5.0 $dm^3$. What mass of this gas will occupy 250 ml at the same temperature and pressure ?

(A) 2.5 gm (B) 250 mg (C) 5.0 gm (D) 100 gm

**IG0203**

**15.** A glass bulb contains 10 gm $O_2$ gas. Now, the bulb is completely evacuated and re-filled with $CH_4$ gas at the same temperature and pressure. The mass of $CH_4$ gas used in refilling is :

(A) 20 gm (B) 10 gm (C) 5 gm (D) 2.5 gm

**IG0204**

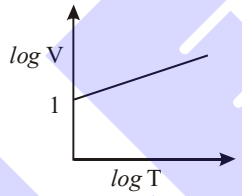
## IDEAL GAS

**16.** A vessel contains mono atomic 'He' at 1 bar and 300 K, determine its number density -

(A) $2.4 \times 10^{25}\ m^{-3}$ (B) $6.8 \times 10^{23}\ m^{-3}$ (C) $4.8 \times 10^{26}\ m^{-3}$ (D) $9.2 \times 10^{27}\ m^{-3}$

**IG0069**

**17.** At constant pressure of 0.82 atm ; log V vs log T is ploted as shown in-
Then number of moles present in experiment -




(A) 1 (B) 10 (C) 100 (D) 0.1

**IG0062**

**18.** A gas is found to have a formula $[CO]_x$. If its vapour density is 70 the value of x is

(A) 2.5 (B) 3.0 (C) 5.0 (D) 6.0

**IG0066**

**19.** A container when is empty weighs 50 gm. After certain liquid of density 25 gm/$dm^3$ is filled its mass becomes equal to 100 gm. The volume of the container will be :

(A) 0.25 $dm^3$ (B) 0.5 $dm^3$ (C) 1 $dm^3$ (D) 2 $dm^3$

**IG0068**

**20.** A gas at a pressure of 5.0 atm is heated from 0°C to 546°C and simultaneously compressed to one-third of its orginal volume. Hence final pressure is

(A) 10 atm (B) 30 atm (C) 45 atm (D) 5 atm

**IG0063**

**21.** A flask containing air (open to the atmosphere) is heated from 300 K to 500 K. Then % of air escaped to the atmosphere is -

(A) 20.0 (B) 40 (C) 60 (D) 80

**IG0064**

**22.** A rigid container containing 10 gm gas at some pressure and temperature. The gas has been allowed to escape from the container due to which pressure of the gas becomes half of its initial pressure and temperature become $(2/3)^{rd}$ of its initial. The mass of gas (in gms) escaped is

(A) 7.5 (B) 1.5 (C) 2.5 (D) 3.5

**IG0070**

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**23.** The density of gas A is twice that of B at the same temperature the molecular weight of gas B is thrice that of A. The ratio of pressure of gas A and gas B will be

(A) 1 : 6 (B) 7 : 8 (C) 6 : 1 (D) 1 : 4

**IG0071**

**24.** In a rigid container $NH_3$ is kept at certain temperature, if on doubling the temperature it is completely dissociated into $N_2$ and $H_2$. Find final pressure to initial pressure ratio :

(A) 4 (B) 2 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

**IG0072**

**25.** 4.0 g of argon has pressure P and temperature T K in a vessel. On keeping the vessel at 50° higher temperature, 0.8 g of argon was given out to maintain the pressure P. The original temperature was :

(A) 73 K (B) 100 K (C) 200 K (D) 510 K

**IG0075**

## DALTON'S LAW OF PARTIAL PRESSURE

**26.** The total pressure exerted by a number nonreacting gases is equal to the sum of partial pressure of the gases under the same conditions is known as :

(A) Boyle's law (B) Charle's law (C) Avogadro's law (D) Dalton's law

**IG0076**

**27.** Dalton's law cannot be applied for which gaseous mixture at normal temperatures:

(A) $O_2$ and $N_2$ (B) $NH_3$ and HCl (C) He and $N_2$ (D) $CO_2$ and $O_2$

**IG0077**

**28.** Equal masses of methane and oxygen are mixed in an empty container at 25°C. The fraction of the total pressure exerted by oxygen is - **[AIEEE-2008]**

(A) 2/3 (B) $\frac{1}{3}\times\frac{273}{298}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

**IG0078**

**29.** A closed vessel contains helium and ozone at a pressure of P atm. The ratio of He and oxygen atoms is 1 : 1. If helium is removed from the vessel, the pressure of the system will reduce to :

(A) 0.5 P atm (B) 0.75 P atm (C) 0.25 P atm (D) 0.33 P atm

**IG0079**

**30.** At constant temperature 200 $cm^3$ of $N_2$ at 720 mm and 400 $cm^3$ of $O_2$ at 750 mm pressure are put together in a one litre flask. The final pressure of mixture is

(A) 111 mm (B) 222 mm (C) 333 mm (D) 444 mm

**IG0080**

**31.** If saturated vapours are compressed slowly (temperature remaining constant) to half the initial volume, the vapour pressure will :

(A) Become four times (B) become doubled

(C) Remain unchanged (D) Become half

**IG0081**

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**32.** A box of 1L capacity is divided into two equal compartments by a thin partiion which are filled with 2g $H_2$ and 16g $CH_4$ respectively. The pressure in each compartment is recorded as P atm. The total pressure when partition is removed will be :

(A) P (B) 2P (C) P/2 (D) P/4

**IG0082**

**33.** A vessel has $N_2$ gas and water vapours at a total pressure of 1 atm. The vapour pressure of water is 0.3 atm. The contents of this vessel are transferred to another vessel having one third of the capacity of original volume, completely at the same temperature the total pressure of this system in the new vessel is -

(A) 3.0 atm (B) 1 atm (C) 3.33 atm (D) 2.4 atm

**IG0083**

## GRAHAM'S LAW OF DIFFUSION OR EFFUSION

**34.** Which gas effuses fastest under identical conditions -

(A) $N_2$ (B) $O_2$ (C) $Cl_2$ (D) $CH_4$

**IG0084**

**35.** The rate of diffusion of methane at a given temperature is twice that of a gas X. The molecular weight of X is :

(A) 64 (B) 32 (C) 4.0 (D) 8.0

**IG0085**

**36.** The rate of diffusion of two gases A and B is in the ratio of 1 : 4 and that of B and C in the ratio of 1 : 3 the rate of diffusion of C with respect to A is -

(A) $\frac{1}{12}$ (B) 12 (C) 6 (D) 4

**IG0086**

**37.** A gas X diffuses three times faster than another gas Y the ratio of their densities i.e., $D_x : D_y$ is

(A) 1/3 (B) 1/9 (C) 1/6 (D)1/12

**IG0087**

**38.** Rate of diffusion of a gas is :

(A) directly proportional to its density

(B) directly proportional to its molecular weight

(C) directly proportional to the square of its molecular weight

(D) inversely proportional to the square root of its molecular weight

**IG0088**

**39.** Since the atomic weights of carbon nitrogen and oxygen are 12, 14 and 16 respectively, among the following pairs of gases, the pair that will diffuse at the same rate is :

(A) $CO_2$ and $N_2O$ (B) $CO_2$ and $N_2O_3$

(C) $CO_2$ and CO (D) $CO_2$ and NO

**IG0089**

**40.** If the four tubes of a car are filled to the same pressure with $N_2$, $O_2$, $H_2$ and He separately then which one will be filled first :

(A) $N_2$ (B) $O_2$ (C) $H_2$ (D) He

**IG0090**

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**41.** The increasing order of effusion among the gases, $H_2$, $O_2$, $NH_3$ and $CO_2$ is –

(A) $H_2$, $CO_2$, $NH_3$, $O_2$ (B) $H_2$, $NH_3$, $O_2$, $CO_2$

(C) $H_2$, $O_2$, $NH_3$, $CO_2$ (D) $CO_2$, $O_2$, $NH_3$, $H_2$

**IG0091**

**42.** The rate of diffusion of hydrogen is about–

(A) $\frac{1}{2}$ that of Helium (B) 1.4 times that of Helium

(C) twice that of H (D) Four times that of Helium

**IG0092**

**43.** Under identical experiment conditions which of the following pairs of gases will be most easy to separate by using effusion process -

(A) $H_2$ and $T_2$ (B) $SO_2$ and $SO_3$ (C) $NH_3$ and $CH_4$ (D) $U^{235}O_2$ and $U^{238}O_2$

**IG0095**

**44.** Certain volume of He gas takes 10 sec for its diffusion, how much time will be taken by $CH_4$ gas to disffuse it's same volume under identical conditions -

(A) 5 sec (B) 10 sec (C) 20 sec (D) 40 sec

**IG0096**

**45.** A football bladder contains equimolar proportions of $H_2$ and $O_2$. The composition by mass of the mixture effusing out of punctured football is in the ratio ($H_2$ : $O_2$)

(A) 1 : 4 (B) $2\sqrt{2}$ : 1 (C) 1 : $2\sqrt{2}$ (D) 4 : 1

**IG0097**

**46.** Consider the following pairs of gases A and B.

| | A | B |
|---|---|---|
| (a) | CO | $N_2$ |
| (b) | $O_2$ | $O_3$ |
| (c) | $^{235}UF_6$ | $^{238}UF_6$ |

Relative rates of effusion of gases A to B under similar condition is in the order:

(A) a < b < c (B) a < c < b (C) a > b > c (D) a > c > b

**IG0098**

**47.** A 4 : 1 molar mixture of He & $CH_4$ kept in a vessel at 20 bar pressure. Due to a hole in the vessel, gas mixture leaks out. What is the composition of mixture effusing out initially -

(A) 8 : 1 (B) 4 : 1 (C) 1 : 4 (D) 4 : 3

**IG0100**

**48.** Calculate the ratio of rate of effusion of $O_2$ and $H_2$ from a container containing 16gm $O_2$ and 2gm $H_2$

(A) 1 : 8 (B) 8 : 1 (C) 1 : 4 (D) 4 : 1

**IG0101**

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## KINETIC THEORY OF GASES

**49.** At STP, the order of root mean square speed of molecules $H_2$, $N_2$, $O_2$ and HBr is :

(A) $H_2 > N_2 > O_2 > HBr$ (B) $HBr > O_2 > N_2 > H_2$

(C) $HBr > H_2 > O_2 > N_2$ (D) $N_2 > O_2 > H_2 > HBr$

**IG0104**

**50.** Four particles have speed 2, 3, 4 and 5 cm/s respectively. Their rms speed is :

(A) 3.5 cm/s (B) $\left(\frac{27}{2}\right)$ cm/s (C) $\sqrt{54}$ cm/s (D) $\left(\frac{\sqrt{54}}{2}\right)$ cm/s

**IG0105**

**51.** A flask has 10 molecules out of which four molecules are moving at $7ms^{-1}$ & the remaining are moving at same speed of X $ms^{-1}$. If Urms of the gas is 5 $ms^{-1}$. The value of 'X' will be

(A) 5 (B) 3 (C) 9 (D) 16

**IG0106**

**52.** If the average velocity of $N_2$ molecules is 0.3 m/sec. at $27^0$C, then the velocity of 0.6 m/sec will take place at :

(A) 273 K (B) 927 K (C) 1000 K (D) 1200 K

**IG0107**

**53.** Which of the following expression does not give root mean square velocity –

(A) $\left(\frac{3RT}{M_w}\right)^{\frac{1}{2}}$ (B) $\left(\frac{3P}{DM_w}\right)^{\frac{1}{2}}$ (C) $\left(\frac{3P}{D}\right)^{\frac{1}{2}}$ (D) $\left(\frac{3PV}{nM_w}\right)^{\frac{1}{2}}$

**IG0108**

**54.** Which one of the following gases would have the highest R.M.S. velocity at $25^0$C ?

(A) Oxygen (B) Carbon dioxide (C) Sulphur dioxide (D) Carbon monoxide

**IG0109**

**55.** At what temperature would the rms speed of a gas molecule have twice its value at 100°C ?

(A) 4192 K (B) 1492 K (C) 9142 K (D) 2491 K

**IG0110**

**56.** Temperature at which most probable speed of $O_2$ becomes equal to root mean square speed of $N_2$ is **[Given : $N_2$ at 427°C]**

(A) 732 K (B) 1200 K (C) 927 K (D) 800 K

**IG0112**

**57.** The density ratio of $O_2$ and $H_2$ is 16 : 1. The ratio of their $U_{rms}$ is :-

(A) 4 : 1 (B) 16 : 1 (C) 1 : 4 (D) 1 : 16

**IG0113**

**58.** The av. K.E./mole of an ideal monoatomic gas at $27^0$C is

(A) 900 cal (B) 1800 cal (C) 300 cal (D) None

**IG0115**

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**59.** The average kinetic energy of an ideal gas per molecule in SI units at $25^0$C will be :
(A) $6.17 \times 10^{-21}$ kJ (B) $6.17 \times 10^{-21}$ J (C) $6.17 \times 10^{-20}$ J (D) $7.16 \times 10^{-20}$ J

**IG0116**

**60.** Average K.E. of $CO_2$ at $27^0$C is E. The average kinetic energy of $N_2$ at the same temperature will be
(A) E (B) 22E (C) E/22 (D) $E/\sqrt{2}$

**IG0118**

**61.** If a gas expands at constant temperature then :
(A) No. of gaseous molecule decreases (B) kinetic energy of molecule decreases
(C) K.E. remain same (D) K.E. increases

**IG0119**

**62.** Total translational kinetic energy possessed by 8 gm methane at 273°C
(R = 2 Cal/mol-K)
(A) 819 calorie (B) 409.5 calorie (C) 1638 calorie (D) None of these

**IG0120**

**63.** Ideal gas equation in terms of K.E. per unit volume E, is-

(A) $P = \frac{3}{2}RT$ (B) $P = \frac{2}{3}E$ (C) $P = \frac{2}{3}RT$ (D) $P = \frac{3}{2}RT$

**IG0121**

**64.** According to the kinetic theory of gases, in an ideal gas, between two successive collisions a gas molecule travels -
**[AIEEE-2003]**
(A) In a straight line path (B) with an accelerated velocity
(C) In a circular path (D) In a wavy path

**IG0122**

**65.** As the temperature is raised from 20°C to 40°C, the average kinetic energy of neon atoms changes by factor of which of the following ?
**[AIEEE-2004]**
(A) 1/2 (B) $\sqrt{(313/293)}$ (C) 313/293 (D) 2

**IG0124**

**66.** Three gases A, B and C are at same temperature. If their r.m.s. speed are in the ratio $1 : \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{3}}$ then their molar masses will be in the ratio :
(A) 1 : 2 : 3 (B) 3 : 2 : 1 (C) $1 : \sqrt{2} : \sqrt{3}$ (D) $\sqrt{3} : \sqrt{2} : 1$

**IG0125**

**67.** The molecular velocity of any gas is :- **[AIEEE-2011]**
(A) inversely proportional to the square root of temperature
(B) inversely proportional to absolute temperature
(C) directly proportional to square of temperature
(D) directly proportional to square root of temperature

**IG0142**



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**68.** $\alpha$, v and u represent most probable velocity, average velocity and root mean square velocity respectively of a gas at a particular temperature. The correct order among the following is -**[JEE(Main)-2012]**

(A) $\alpha > u > v$ (B) $v > u > \alpha$ (C) $u > v > \alpha$ (D) $u > \alpha > v$

**IG0143**

**69.** The relationship among most probable velocity, average velocity and root mean square velocity is respectively :- **[JEE(Main-online)-2012]**

(A) $\sqrt{2} : \sqrt{8/\pi} : \sqrt{3}$ (B) $\sqrt{2} : \sqrt{3} : \sqrt{8/\pi}$ (C) $\sqrt{3} : \sqrt{8/\pi} : \sqrt{2}$ (D) $\sqrt{8/\pi} : \sqrt{3} : \sqrt{2}$

**IG0146**

## COLLISION PARAMETERS

**70.** The following gases are present under similar condition of T, P & V. The longest mean free path stands for

(A) $H_2$ (B) $N_2$ (C) $O_2$ (D) $Cl_2$

**IG0126**

**71.** At constant volume, $Z_{11}$ is directly proportional to -

(A) $\sqrt{P}$ (B) P (C) $T^2$ (D) T

**IG0127**

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## EXERCISE # O-II

**Single Correct :**

**1.** A gaseous reaction,

$3A \longrightarrow 2B$

is carried out in a 0.082 litre closed container initially containing 1 mole of gas A. After sufficient time a curve of P (atm) vs T (K) is plotted and the angle with x-axis was found to be 42.95°. The degree of association of gas A is **[Given : tan 42.95 = 0.8]**

(A) 0.4 (B) 0.6 (C) 0.5 (D) 0.8

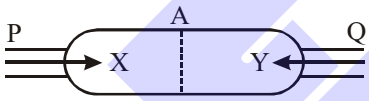
**IG0074**

**2.** Under identical conditions of pressure and temperature. 4 L of gaseous mixture ($H_2$ and $CH_4$) effuses through a hole in 5min whereas 4 L of a gas X of molecular mass 36 takes to 10 min to effuse through the same hole. The mole ratio of $H_2$ : $CH_4$, in the mixture is -

(A) 1 : 2 (B) 2 : 1 (C) 2 : 3 (D) 1 : 1

**IG0093**

**3.** 3 mole of gas "X" and 2 moles of gas "Y enters from end "P" and "Q" of the cylinder respectively. The cylinder has the area of cross-section A, shown as under -




The length of the cylinder is 150 cm. The gas "X" intermixes with gas "Y" at the point A. If the molecular weight of the gases X and Y is 20 and 80 respectively, then what will be the distance of point A from Q?

(A) 75cm (B) 50cm (C) 37.5 cm (D) 90 cm

**IG0094**

**4.** For the reaction

$2NH_3(g) \longrightarrow N_2(g) + 3H_2(g)$,

what is the % of $NH_3$ converted if the mixture diffuses twice as fast as that of $SO_2$ under similar conditions.

(A) 3.125 % (B) 31.25 % (C) 6.25 % (D) 62.5 %

**IG0099**

**5.** Two flasks X and Y have capacity 1L and 2L respectively and each of them contains 1 mole of a gas. The temperature of the flask are so adjusted that average speed of molecules in X is twice as those in Y. The pressure in flask X would be

(A) same as that in Y (B) half of that in Y (C) twice of that in Y (D) 8 times of that in Y

**IG0111**

**6.** Which of the gas have highest fraction of molecules at 27ºC in most probable speed region -

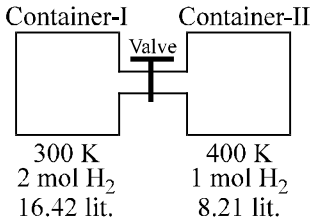
(A) $H_2$ (B) $N_2$ (C) $O_2$ (D) $CO_2$

**IG0114**

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**One or more may be correct :**

**7.** If both the containers are maintained at constant temperature (R = 0.0821 L-atm/K-mol) Select the correct option(s) :




(A) Pressure in container-I is 3 atm before opening the valve.

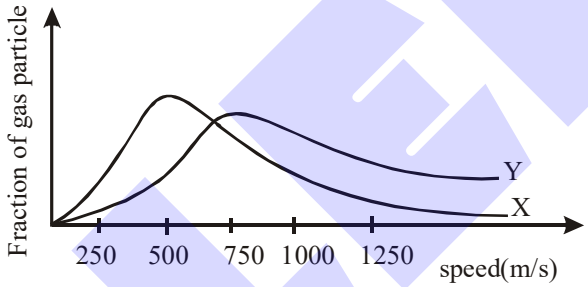
(B) Pressure after opening the valve is 3.57 atm.

(C) Moles in each container are same after opening the valve.

(D) Pressure in each container are same after opening the valve.

**IG0129**

**8.** The graph below shows the distribution of molecular speed of two ideal gases X and Y at 200K. on the basis of the below graph identify the correct statements -




(A) If gas X is methane, then gas Y can be $CO_2$

(B) Fraction of molecules of X must be greater than Y in a particular range of speed at 200K

(C) Under identical conditions rate of effusion of Y is greater than that of X

(D) The molar kinetic energy of gas X at 200K is equal to the molar kinetic energy of Y at 200K

**IG0130**

**9.** Identify the correct statements when a fixed amount of ideal gas is heated in a container fitted with a movable piston always operating at constant pressure.

(A) Average distance travelled between successive collisions will decreases.

(B) Collisions frequency increases since speed of the molecules increases with increase in temperature.

(C) Average relative speed of approach remains unaffected.

(D) Average angle of approach remains unaffected.

**IG0131**

**10.** Choose the correct statement(s) among the following

(A) Average molecular speed of gases increases with decrease in fraction of molecules moving slowly

(B) Rate of effusion of gases increases with increase in collision frequency at constant volume.

(C) Rate of effusion is inversely proportional to molecular weight of gas

(D) Mean free path does not change with change in temperature at constant pressure.

**IG0132**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\V4-Ideal-Reas-Atomic-Chemical Eq\Eng\1. Ideal\02. Ex.p65

**11.** Which of the following quantities is the same for all ideal gases at the same temperature :
(A) The kinetic energy of 1 mol
(B) The kinetic energy of 1 g
(C) The number of molecules in 1 mol
(D) The number of molecules in 1 g

**IG0133**

**12.** Choose the correct statement(s) among the following -
(A) The mean free path ($\lambda$) of gaseous molecules is directly proportional to temperature of gas at constant volume
(B) The mean free path ($\lambda$) of gaseous molecules is inversely proportional to pressure of gas at constant volume
(C) The mean free path ($\lambda$) of gaseous molecules is directly proportional to volume of gas at constant T
(D) The mean free path ($\lambda$) of gaseous molecules is directly proportional to volume of gas at constant P

**IG0134**

**13.** Which statement is/are correct for postulates of kinetic theory of gases -
(A) Gases are composed of molecules whose size is negligible compared with the average distance between them
(B) Molecules moves randomly in straight lines in all directions and at various speeds.
(C) When molecules collide with one another the collisions are elastic. In an elastic collision the loss of kinetic energy takes place
(D) The average kinetic energy of a molecule is proportional to the absolute temperature.

**IG0135**

**Paragraph for Question 14 to 16**

**The constant motion and high velocities of gas particles lead to some important practical consequences. One such consequence is that as minimum rapidly when they come in contact. The mixing of different gases by random molecular motion and with frequent collisions is called diffusion A similar process in which gas molecules escape through a tiny hole into a vaccume is called effusion.**

**14.** Helium gas at 1 atm and $SO_2$ at 2 atm pressure, temperature being the same, are released separately at the same moment into 1 m long evacuated tubes of equal diameters. If helium reaches the other end of the tube in t sec, what distance $SO_2$ would traverse in the same time interval in the other tube ?
(A) 25 cm (B) 50 cm (C) 60 cm (D) 75 cm

**IG0136**

**15.** 4 g of $H_2$ effused through a pinhole in 10 sec at constant temperature and pressure. The amount of oxygen effused in the same time interval and at the same conditions of temperature and pressure would be :
(A) 4 g (B) 8 g (C) 16 g (D) 32 g

**IG0137**

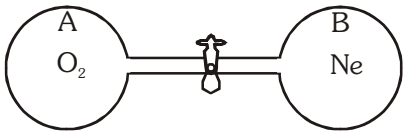
**16.** For 10 min. each at 27°C, from two identical bulbs helium and an unknown gas X at equal pressures are leaked into a common vessel of 3 L capacity. The resulting pressure is 4.1 atm and the mixture contains 0.4 mol of helium. The molar mass of gas X is :
(A) 16 (B) 32 (C) 64 (D) None of these

**IG0138**

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Ideal-Reas-Atomic-Chemical Eq\Eng\1. Ideal\02. Ex.p65

**Paragraph for Question 17 to 18**

**Initially, flask A contained oxygen gas at 27°C and 950 mm of Hg, and flask B contained neon gas at 27°C and 900 mm. Finally, the two flasks were joined by means of a narrow tube of negligible volume equipped with a stopcork and gases were allowed to mixup freely. The final pressure in the combined system was found to be 910 mm of Hg.**




**17.** What is the correct relationship between volumes of the two flasks ?

(A) $V_B = 3V_A$ (B) $V_B = 4V_A$ (C) $V_B = 5V_A$ (D) $V_B = 4.5V_A$

**IG0139**

**18.** How many moles of gas are present in flask A in the final condition, if volume of flask B is 304 litre ? ($R = 0.08$ atm L $mol^{-1}$ $K^{-1}$)

(A) 7.58 (B) 3.79 (C) 15.16 (D) None of these

**IG0139**

**TABLE TYPE QUESTION :**

| **Column-I (Gases at different conditions)** | **Column-II (Value of speed (m/s)** | **Column-III** |
|---|---|---|
| (A) $CH_4$ at 27°C | (P) $U_{rms} = 342.5$ | (I) Molar K.E. = 3750 Joule |
| (B) $SO_2$ at 27°C | (Q) $U_{mp} = 1100$ | (II) Molar K.E. = 5000 Joule |
| (C) 4 gm He at 1 atm and 24.6 litre | (R) $U_{rms} = 685$ | (III) Average K.E. per gram = 937.5 Joule |
| (D) 32 gm $O_2$ at 127°C | (S) $U_{rms} = 550$ | (IV) Average K.E. per gram = 234 Joule. |

**(Given :** $R = \frac{25}{3}\frac{J}{mol \times K}$, $\sqrt{30} = 5.48$, $\sqrt{\frac{2}{3}} = 0.8$, $\sqrt{5} = 2.2$**)**

**19.** Which of the following is correct

(A) A ; R ; 4 (B) A ; R ; 2 (C) A ; Q ; 4 (D) B ; R ; 1

**IG0140**

**20.** Which of the following is correct

(A) B ; Q ; 1 (B) B ; Q ; 2 (C) B ; P ; 1 (D) C ; S ; 3

**IG0140**

**21.** Which of the following is correct

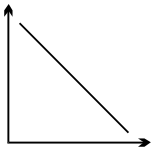
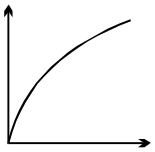
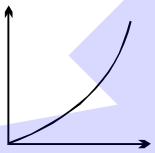
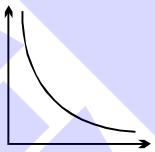
(A) C ; Q ; 3 (B) D ; R ; 2 (C) D ; S ; 1 (D) D ; S ; 4

**IG0140**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\V4-Ideal-Reas-Atomic-Chemical Eq\Eng\1. Ideal\02. Ex.p65

**MATCH THE COLUMN:**

**22.** Match the entries in column I with entries in Column II and then pick out correct options.

| | **Column I** | | **Column II** |
|---|---|---|---|
| (A) | $\frac{1}{V^2}$ vs P for ideal gas at constant T and n. | (P) |  |
| (B) | V vs $\frac{1}{T}$ for ideal gas at constant P and n | (Q) |  |
| (C) | log P vs log V for ideal gas at constant T and n. | (R) |  |
| (D) | V vs $\frac{1}{P^2}$ for ideal gas at constant T and n. | (S) |  |

**IG0141**

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Ideal-Reas-Atomic-Chemical Eq\Eng\1. Ideal\02. Ex.p65

## *EXERCISE # JEE-MAINS*

**1.** Which one of the following is the wrong assumption of kinetic theory of gases ?

**[JEE(Main-online)-2013]**

(1) All the molecules move in straight line between collision and with same velocity.

(2) Molecules are separated by great distances compared to their sizes.

(3) Pressure is the result of elastic collision of molecules with the container's wall.

(4) Momentum and energy always remain conserved.

**IG0147**

**2.** By how many folds the temperature of a gas would increase when the root mean square velocity of the gas molecules in a container of fixed volume is increased from $5 \times 10^4$ cm/s to $10 \times 10^4$ cm/s ?

**[JEE(Main-online)-2013]**

(1) Four (2) three (3) Two (4) Six

**IG0148**

**3.** A gaseous compound of nitrogen and hydrogen contains 12.5%(by mass) of hydrogen. The density of the compound relative to hydrogen is 16. The molecular formula of the compound is :

**[JEE(Main-online)-2014]**

(1) $NH_2$ (2) $NH_3$ (3) $N_3H$ (4) $N_2H_4$

**IG0150**

**4.** The initial volume of a gas cylinder is 750.0 mL. If the pressure of gas inside the cylinder changes from 840.0 mm Hg to 360.0 mm Hg, the final volume the gas will be **[JEE(Main-online)-2014]**

(1)1.750 L (2) 7.50 L (3) 3.60 L (4) 4.032 L

**IG0151**

**5.** The temperature at which oxygen molecules have the same root mean square speed as helium atoms have at 300 K is : **(Atomic masses : He = 4 u, O = 16 u)** **[JEE(Main-online)-2014]**

(1) 1200 K (2) 600 K (3) 300 K (4) 2400 K

**IG0152**

**6.** Which of the following is not an assumption of the kinetic theory of gases ?

(1) Gas particles have negligible volume. **[JEE-Mains (online)-2015]**

(2) A gas consists of many identical particles which are in continual motion.

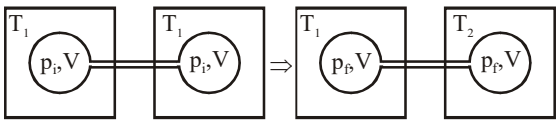
(3) At high pressure, gas particles are difficult to compress.

(4) Collisions of gas particles are perfectly elastic.

**IG0153**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\V4-Ideal-Reas-Atomic-Chemical Eq\Eng\1. Ideal\02. Ex.p65

**7.** Two closed bulbs of equal volume(V) containing an ideal gas initially at pressure $p_i$ and temperature $T_1$ are connected through a narrow tube of negligible volume as shown in the figure below. The temperature of one of the bulbs is then raised to $T_2$. The final pressure $p_f$ is :- **[JEE-Mains-2016]**




(1) $2p_i\left(\frac{T_1T_2}{T_1+T_2}\right)$ (2) $p_i\left(\frac{T_1T_2}{T_1+T_2}\right)$ (3) $2p_i\left(\frac{T_1}{T_1+T_2}\right)$ (4) $2p_i\left(\frac{T_2}{T_1+T_2}\right)$

**IG0154**

**8.** At 300 K, the density of a certain gaseous molecule at 2 bar is double to that of dinitrogen ($N_2$) at 4 bar. The molar mass of gaseous molecule is:- **[JEE-Mains-2017(ONLINE)]**

(1) 28 g $mol^{-1}$ (2) 56 g $mol^{-1}$ (3) 224 g $mol^{-1}$ (4) 112 g $mol^{-1}$

**IG0155**

**9.** Assuming ideal gas behaviour, the ratio of density of ammonia to that of hydrogen chloride at same temperature and pressure is : **(Atomic wt. of Cl = 35.5 u)** **[JEE-Mains-2018(ONLINE)]**

(1) 0.64 (2) 1.64 (3) 1.46 (4) 0.46

**IG0156**

**10.** Points I, II and III in the following plot respectively correspond to.

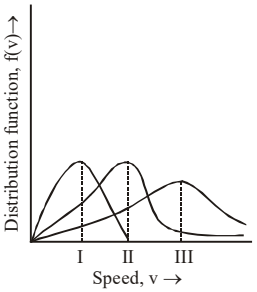
($V_{mp}$ : most probable velocity) **[JEE-Mains-2019(ONLINE)]**

(1) $V_{mp}$ of $N_2$ (300K); $V_{mp}$ of $H_2$(300K); $V_{mp}$ of $O_2$(400K)

(2) $V_{mp}$ of $H_2$ (300K); $V_{mp}$ of $N_2$(300K); $V_{mp}$ of $O_2$(400K)

(3) $V_{mp}$ of $O_2$ (400K); $V_{mp}$ of $N_2$(300K); $V_{mp}$ of $H_2$(300K)

(4) $V_{mp}$ of $N_2$ (300K); $V_{mp}$ of $O_2$(400K); $V_{mp}$ of $H_2$(300K)



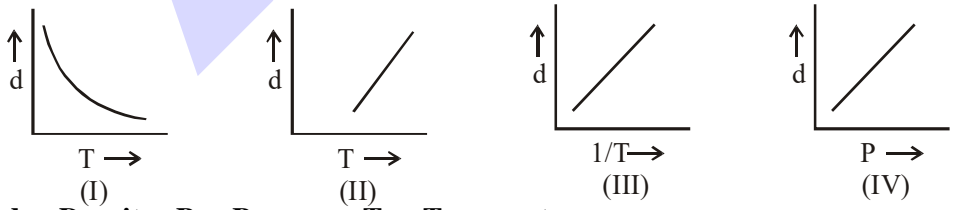

**IG0157**

**11.** An open vessel at 27°C is heated until two fifth of the air (assumed as an ideal gas) in it has escaped from the vessel. Assuming that the volume of the vessel remains constant, the temperature at which the vessel has been heated is : **[JEE-Mains-2019(ONLINE)]**

(1) 750°C (2) 500°C (3) 750 K (4) 500 K

**IG0159**

**12.** Which one of the following graphs is **not correct** for ideal gas ?




**d = Density, P = Pressure, T = Temperature**

(1) II (2) III (3) I (4) IV

**[JEE(Main-online)-2020]**

**IG0205**

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Ideal-Reas-Atomic-Chemical Eq\Eng\1. Ideal\02. Ex.p65

**13.** A mixture of one mole each of $H_2$, He and $O_2$ each are enclosed in a cylinder of volume V at temperature T. If the partial pressure of $H_2$ is 2 atm, the total pressure of the gases in the cylinder is :

(1) 14 atm (2) 22 atm (3) 6 atm (4) 38 atm

**[JEE(Main-online)-2020]**

**IG0206**

**14.** A spherical balloon of radius 3 cm containing helium gas has a pressure of $48 \times 10^{-3}$ bar. At the same temperature, the pressure, of a spherical balloon of radius 12 cm containing the same amount of gas will be_____$\times 10^{-6}$ bar. **[JEE(Main-online)-2020]**

**IG0207**

**15.** The pressure exerted by a non-reactive gaseous mixture of 6.4 g of methane and 8.8 g of carbon dioxide in a 10 L vessel at 27°C is _____ kPa.

(Round off to the Nearest Integer).

**[Assume gases are ideal, R = 8.314 J $mol^{-1}$ $K^{-1}$**

**Atomic masses : C : 12.0 u, H : 1.0 u, O : 16.0 u]** **[JEE(Main-online)-2021]**

**IG0208**

**16.** An LPG cylinder contains gas at a pressure of 300 kPa at 27°C. The cylinder can withstand the pressure of $1.2 \times 10^6$ Pa. The room in which the cylinder is kept catches fire. The minimum temperature at which the bursting of cylinder will take place is _____ °C. (Nearest integer)

**[JEE(Main-online)-2021]**

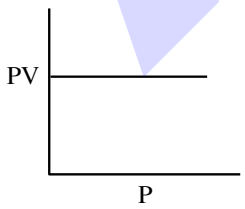
**IG0209**

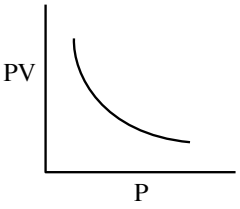
**17.** A home owner uses $4.00 \times 10^3$ $m^3$ of methane ($CH_4$) gas, (assume $CH_4$ is an ideal gas) in a year to heat his home. Under the pressure of 1.0 atm and 300 K, mass of gas used is $x \times 10^5$ g. The value of x is _____ .

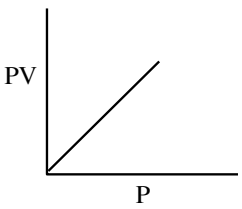
(Nearest integer) **(Given R = 0.083 L atm $K^{-1}$ $mol^{-1}$)** **[JEE(Main-online)-2021]**

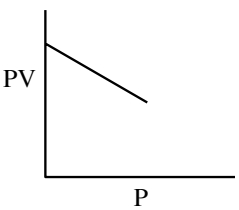
**IG0210**

**18.** Which one of the following is the correct PV vs P plot at constant temperature for an ideal gas ? (P and V stand for pressure and volume of the gas respectively)

(1) 


(2) 


(3) 


(4) 


**[JEE(Main-online)-2021]**

**IG0212**

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\V4-Ideal-Reos-Atomic-Chemical Eq\Eng\1. Ideal\02. Ex.p65

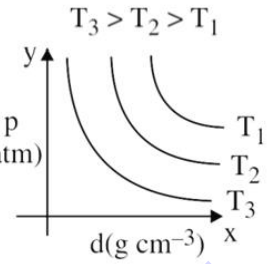
**19.** An empty LPG cylinder weighs 14.8 kg. When full, it weighs 29.0 kg and shows a pressure of 3.47 atm. In the course of use at ambient temperature, the mass of the cylinder is reduced to 23.0 kg. The final pressure inside of the cylinder is _____atm. (Nearest integer)
**(Assume LPG of be an ideal gas)** **[JEE(Main-online)-2021]**
**IG0213**

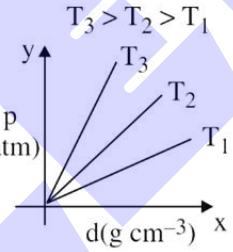
**20.** At 300 K, a sample of 3.0 g of gas A occupies the same volume as 0.2 g of hydrogen at 200 K at the same pressure. The molar mass of gas A is____ g $mol^{-1}$ (nearest integer) Assume that the behaviour of gases as ideal. (Given: The molar mass of hydrogen ($H_2$) gas is 2.0 g $mol^{-1}$)
**[JEE(Main-online)-2022]**
**IG0214**

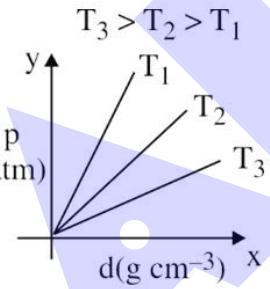
**21.** A rigid nitrogen tank stored inside a laboratory has a pressure of 30 atm at 06:00 am when the temperature is 27 °C. At 03:00 pm, when the temperature is 45°C, the pressure in the tank will be _____atm. [nearest integer] **[JEE(Main-online)-2022]**
**IG0215**

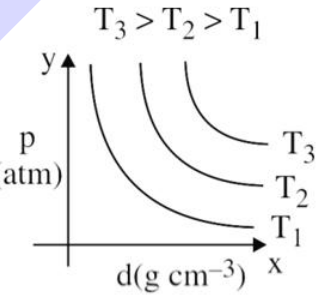
**22.** An evacuated glass vessel weighs 40.0 g when empty, 135.0 g when filled with a liquid of density 0.95 g $mL^{-1}$ and 40.5 g when filled with an ideal gas at 0.82 atm at 250 K. The molar mass of the gas in g $mol^{-1}$ is : **(Given : R = 0.082 L atm $K^{-1}$ $mol^{-1}$)** **[JEE(Main-online)-2022]**
(A) 35 (B) 50 (C) 75 (D) 125
**IG0216**

**23.** Which amongst the given plots is the correct plot for pressure (p) vs density (d) for an ideal gas ?

(A) 


(B) 


(C) 


(D) 


**[JEE(Main-online)-2022]**
**IG0217**

**24.** CNG is an important transportation fuel. When 100 g CNG is mixed with 208 oxygen in vehicles, it leads to the formation of $CO_2$ and $H_2O$ and produces large quantity of heat during this combustion, then the amount of carbon dioxide, produced in grams is_____. [nearest integer]

**[Assume CNG to be methane]** **[JEE(Main-online)-2022]**
**IG0218**

node06\B09C-BD\Kota\JEE(Advanced)\Nurture\Chem\Module\4-Ideal-Real-Atomic-Chemical Eq\Eng\1. Ideal\02. Ex.p65

**25.** A box contains 0.90 g of liquid water in equilibrium with water vapour at 27°C. The equilibrium vapour pressure of water at 27°C 32.0 Torr. When the volume of the box is increased, some of the liquid water evaporates to maintain the equilibrium pressure. If all the liquid water evaporates, then the volume of the box must be____ litre. [nearest integer]

**(Given: R = 0.082 L atm $K^{-1}$ $mol^{-1}$)** **[JEE(Main-online)-2022]**

**(Ignore the volume of the liquid water and assume water vapours behave as an ideal gas.)**

**IG0219**

**26.** Geraniol, a volatile organic compound, is a component of rose oil. The density of the vapour is 0.46 $gL^{-1}$ at 257°C and 100 mm Hg. The molar mass of geraniol is _____ (Nearest Integer)

**[Given R = 0.082 L atm $K^{-1}$ $mol^{-1}$]** **[JEE(Main-online)-2022]**

**IG0220**

**27.** A sealed flask with a capacity of 2 $dm^3$ contains 11 g of propane gas. The flask is so weak that it will burst if the pressure becomes 2 MPa. The minimum temperature at which the flask will burst is _____ °C. [Nearest integer]

(Given: R = 8.3 J $K^{-1}$ $mol^{-1}$. Atomic masses of C and H are 12u and 1u respectively.) (Assume that propane behaves as an ideal gas.) **[JEE(Main-online)-2022]**

**IG0221**

**28.** A mixture of hydrogen and oxygen contains 40% hydrogen by mass when the pressure is 2.2 bar. The partial pressure of hydrogen is_____bar. (Nearest Integer) **[JEE(Main-online)-2022]**

**IG0222**

**29.** 'x' g of molecular oxygen ($O_2$) is mixed with 200 g of neon (Ne). The total pressure of the non-reactive mixture of $O_2$ and Ne in the cylinder is 25 bar. The partial pressure of Ne is 20 bar at the same temperature and volume. The value of 'x' is_____.

**[Given: Molar mass of $O_2$ = 32 g $mol^{-1}$.**

**Molar mass of Ne = 20 g $mol^{-1}$]** **[JEE(Main-online)-2022]**

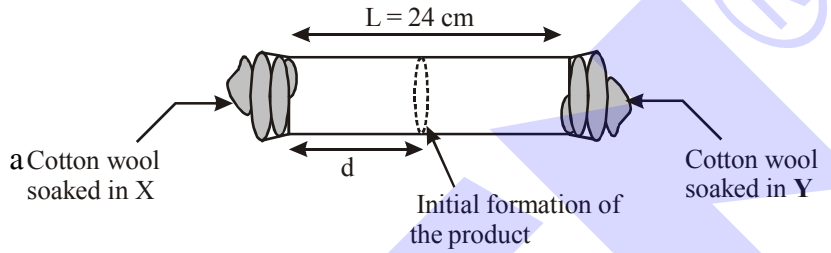
**IG0223**

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## EXERCISE # JEE-ADVANCED

**Paragraph for Question 1 & 2**

**X and Y are two volatile liquids with molar weights of 10 g $mol^{-1}$ and 40 g $mol^{-1}$ respectively. Two cotton plugs, one soaked in X and the other soaked in Y, are simultaneously placed at the ends of a tube of length L = 24 cm, as shown in the figure. The tube is filled with an inert gas at 1 atmosphere pressure and a temperature of 300 K. Vapours of X and Y react to form a product which is first observed at a distance d cm from the plug soaked in X. Take X and Y to have equal molecular diameters and assume ideal behaviour for the inert gas and the two vapours.** **[JEE 2014]**




**1.** The value of d in cm (shown in the figure), as estimated from Graham's law, is -

(A) 8 (B) 12 (C) 16 (D) 20

**IG0162**

**2.** The experimental value of d is found to be smaller than the estimate obtained using Graham's law. This is due to -

(A) Larger mean free path for X as compared to that of Y

(B) Larger mean free path for Y as compared to that of X

(C) Increased collision frequency of Y with the inert gas as compared to that of X with the inert gas

(D) Increased collision frequency of X with the inert gas as compared to that of Y with the inert gas

**IG0163**

**3.** The diffusion coefficient of an ideal gas is proportional to its mean free path and mean speed. The absolute temperature of an ideal gas is increased 4 times and its pressure is increased 2 times. As a result, the diffusion coefficient of this gas increases x times. The value of x is **[JEE 2016]**

**IG0224**

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**4.** A closed tank has two compartments A and B, both filled with oxygen (assumed to be ideal gas). The partition separating the two compartments is fixed and is a perfect heat insulator (Figure 1). If the old partition is replaced by a new partition which can slide and conduct heat but does NOT allow the gas to leak across (Figure 2), the volume (in $m^3$) of the compartment A after the system attains equilibrium is____. **[JEE 2018]**

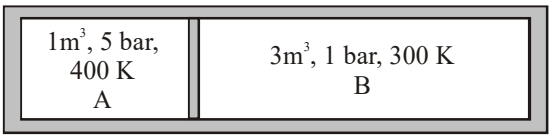



Figure 1

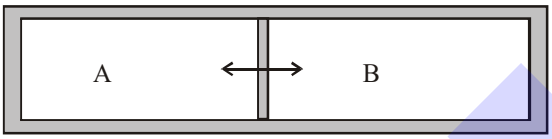



Figure 2

**IG0164**

**5.** Which of the following statement(s) is (are) correct regarding the root mean square speed ($U_{rms}$) and average translational kinetic energy ($\varepsilon_{av}$) of a molecule in a gas at equilibrium ?

(1) $U_{rms}$ is doubled when its temperature is increased four times **[JEE 2019]**

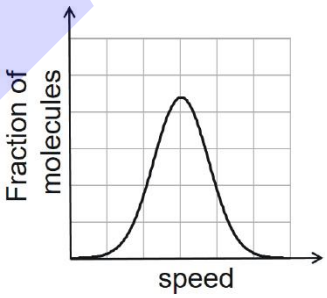
(2) $\varepsilon_{av}$ at a given temperature does not depend on its molecular mass

(3) $U_{rms}$ is inversely proportional to the square root of its molecular mass

(4) $\varepsilon_{av}$ is doubled when its temperature is increased four times

**IG0165**

**6.** If the distribution of molecular speeds of a gas is as per the figure shown below, then the ratio of the most probable, the average and the roots mean square speeds, respectively, is **[JEE 2020]**




(A) 1 : 1 : 1 (B) 1 : 1 : 1.224 (C) 1 : 1.128 : 1.224 (D) 1 : 1.128 : 1

**IG0225**

**7.** Aluminium reacts with sulfuric acid to form aluminium sulfate and hydrogen. What is the volume of hydrogen gas in liters (L) produced at 300 K and 1.0 atm pressure, when 5.4 g of aluminium and 50.0 mL of 5.0 M sulfuric acid are combined for the reaction ?

**(Use molar mass of aluminium as 27.0 g $mol^{-1}$ , R = 0.082 atm L mol-1 $K^{-1}$)** **[JEE 2020]**

**IG0226**

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# ANSWER KEY

## EXERCISE # S-I

| | | | | |
|---|---|---|---|---|
| 1. | (500.00) | 2. (1.00) | 3. (37.50) | 4. (150.00) |
| 5. | (5.46 or 5.47) | 6. (30.00) | 7. (127.00) | 8. (135.00) |
| 9. | (64.28) | 10. (2.68) | 11. (24.60) | 12. (0.60) |
| 13. | (32.00) | 14. (100.00) | 15. (1.73) | 16. (4.00) |
| 17. | (4.00) | 18. (8.00) | 19. (50.00) | 20. (3.98) |
| 21. | (45.00) | 22. (0.06 or 0.07) | 23. i. (15.00) ii. (12.00) | 24. (1.25) |
| 25. | (93.75) | 26. (0.80) | 27. (19.48) | 28. (1.80) |
| 29. | (1.50) | 30. (0.15) | 31. (0.28) | 32. (420.00) |
| 33. | (146.50) | 34. (2.00) | 35. (36.00) | 36. (8.00) |
| 37. | (20.00) | 38. (5.00) | 39. (4.00) | 40. (8.00) |
| 41. | (8.00) | 42. (4.00) | 43. (0.25) | 44. (400.00) |
| 45. | (25.00) | 46. (2.00) | 47. (55.55 or 55.56) | 48. (1.10) |
| 49. | (187.06) | 50. (2.00) | 51. (4.00) | 52. (–173.00) |
| 53. | (1.08 or 1.09) | 54. (2.00) | 55. (0.50) | 56. (1.41 = $\sqrt{2}$) |
| 57. | (0.35 = $\frac{1}{2\sqrt{2}}$) | 58. (4.00) | | |

## EXERCISE # S-II

1. (4.00)
2. ($T_3$ = 360.00, $T_4$ = 90.00, $P_1$ = 5.00, $P_2$ = 1.50, $V_1$ = 16.00)

| | | | |
|---|---|---|---|
| 3. | a. (80.00), b. (1295.00) | 4. i. (6.00), ii. (No) | 5. (627.00) |
| 6. | (2.00) | 7. (6.00) | 8. (2.00) | 9. (380.00) |
| 10. | i. (1.50), ii. (1.25), iii. (0.20 or 0.21) | 11. (7.00) | 12. (4.00) |

13. (8.00)
14. i. (0.012 atm), ii. ($2.50 \times 10^{22}$), iii. (3.60 J), iv. (300.00 J), v. (1 : $\sqrt{2}$), vi. (1 : 2500)

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## EXERCISE # O-I

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | A | B | B | D | D | C | C | B | D | A |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | B | C | A | B | C | A | C | C | D | C |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| Ans. | B | C | C | A | C | D | B | C | C | D |
| Que. | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| Ans. | C | A | D | D | A | B | B | D | A | C |
| Que. | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| Ans. | D | B | A | C | A | B | A | A | A | D |
| Que. | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| Ans. | B | D | B | D | B | B | C | A | B | A |
| Que. | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| Ans. | C | A | B | A | C | A | D | C | A | A |
| Que. | 71 | | | | | | | | | |
| Ans. | A | | | | | | | | | |

## EXERCISE # O-II

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | B | D | C | C | D | D | A,D | C,D | D | A,B |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | A,C | C,D | A,B,D | B | C | C | B | B | A | C |
| Que. | 21 | 22 | | | | | | | | |
| Ans. | A | A-R | B-S | C-P | D-Q | | | | | |

## *EXERCISE # JEE-MAINS*

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans. | 1 | 1 | 4 | 1 | 4 | 3 | 4 | 4 | 4 | 4 |
| Que. | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| Ans. | 4 | 1 | 3 | 750.00 | 150 | 927 | 26 | 1 | 2 | 45 |
| Que. | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | |
| Ans. | 32 | D | B | 143 | 29 | 152 | 1655 | 2 | 80 | |

## *EXERCISE # JEE-ADVANCED*

| Que. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | |
|---|---|---|---|---|---|---|---|---|
| Ans. | C | D | 4 | 2.22 | 1,2,3 | B | 6.15 | |

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# Important Notes

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